

Prof. Dr. Lincy Pyl and Selection Committee
Department of Mechanics of Materials and Constructions (MeMC)
Vrije Universiteit Brussel

Subject: Motivation letter – PhD candidate, Mechanics of Materials and Constructions (M²S)
Introduction video: <https://drive.google.com/file/d/14s-V07z3fopEllnUP8vmMJPMxQW57mg9/view>

Dear Prof. Pyl and Selection Committee,

I am Pooja Awasare, a civil and environmental engineer, and I am applying for the PhD position in Mechanics of Materials and Constructions at VUB. Materials and structures have stayed with me since my civil engineering studies, and I want to take that interest into doctoral research with your group.

During my Diploma and Bachelor's in Civil Engineering in India I spent many hours in the materials laboratory with concrete, steel, soil, aggregates and water. When a test result looked wrong, I did not like to leave it at the number on the page. I wanted to understand what had caused it, and I still work like that: I stay with a question and keep checking until the reason becomes clear.

I later completed a Master's in Environmental Engineering at HTW Dresden. My thesis looked at how construction teams can manage projects and use resources more carefully, and that work pushed me back toward materials and structures. For me they are the heart of building, because sustainable practice only becomes real when we understand how materials behave under load and how structures use them.

In Belgium I worked with a construction partner on circular practices in day-to-day project delivery. That role showed me how waste, reuse and process are handled on real projects, and it also showed me the gap I still want to close on the mechanical side, through laboratory testing and careful analysis.

I have been following MeMC and reading about your group's work, including Prof. Pyl's research on lightweight structures and on the mechanical performance of materials, where experiments are linked with analytical and numerical models, and MeMC's work on resource-efficient and circular building systems. Reading that work made VUB feel concrete to me, not only like a name on a vacancy page.

Thank you for considering my application and for watching my introduction video.

Yours sincerely,

Pooja Awasare